

DRUG REPURPOSING DEMO

Query : 'anemia' → anemia (disease)

Model : HGT

Scoring 100 drug candidates ...

Computing disease similarity ...

OUTPUT 1 — TOP 5 CANDIDATE DRUGS

Disease : anemia (disease) | Model : HGT

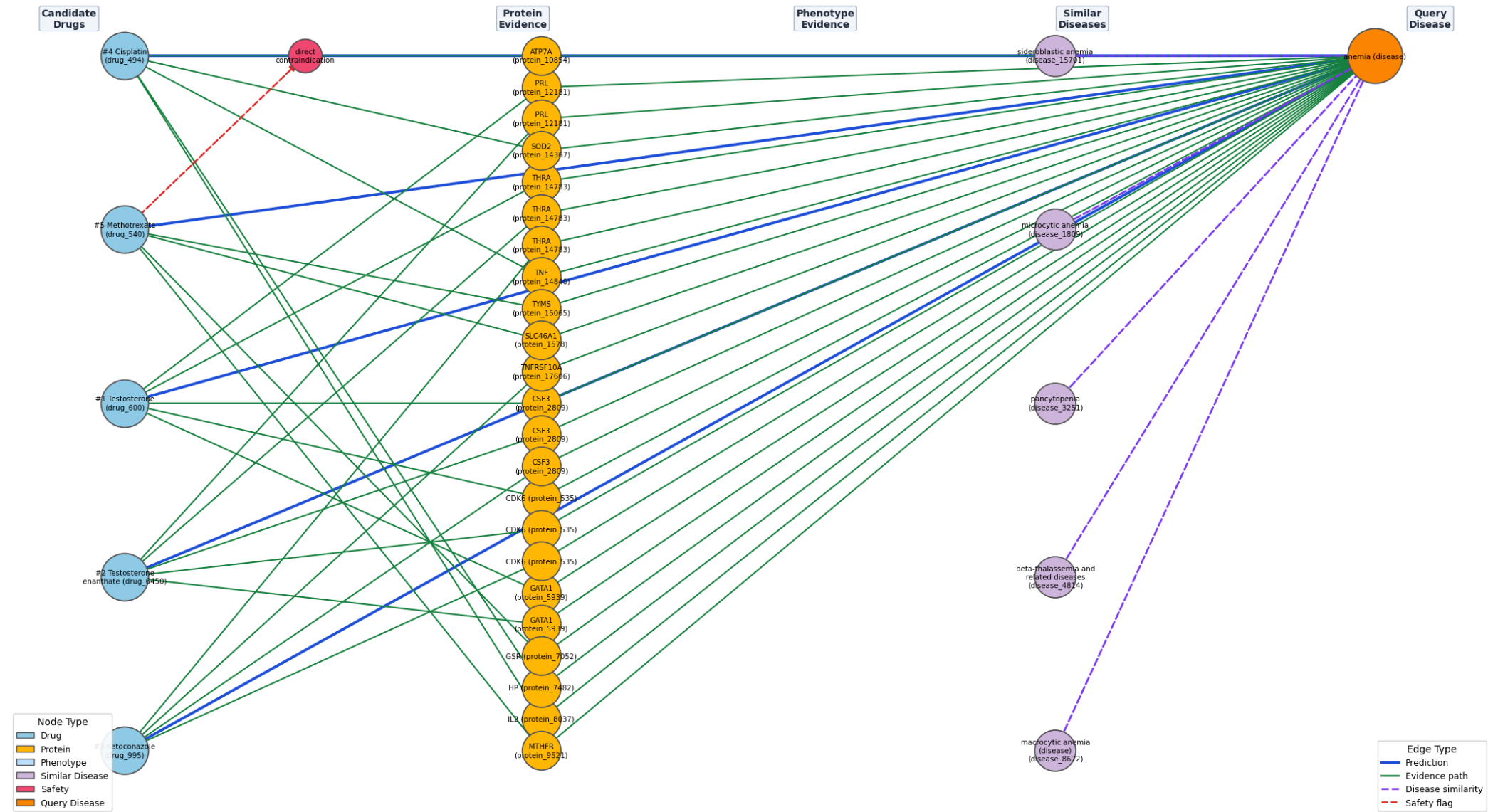
	Drug	Model Score	Sim. Support	Rerank Score	Risk	Evidence	Safety Flags	Known Treatment For
Rank								
1	Testosterone (drug_600)	0.566466	0.0	0.511556	low	5	0	premature menopause, combined pituitary hormone deficiencies, genetic form, pituitary deficiency (+20 more)
2	Testosterone enanthate (drug_6450)	0.562182	0.0	0.509200	low	5	0	premature menopause, trisomy 13, androgen insensitivity syndrome (+3 more)
3	Ketoconazole (drug_995)	0.561272	0.0	0.508700	low	4	0	chromomycosis, white piedra, Cushing syndrome (+11 more)
4	Cisplatin (drug_494)	0.561239	0.0	0.508682	low	6	0	borderline epithelial tumor of ovary, malignant non-dysgerminomatous germ cell tumor of ovary, familial ovarian canc...
5	Methotrexate (drug_540)	0.562198	0.0	0.434209	high	5	1	juvenile idiopathic arthritis, Langerhans cell histiocytosis, non-Hodgkin lymphoma (+15 more)

OUTPUT 2 — TOP 10 SIMILAR DISEASES

	Similar Disease	Similarity Score	Embedding Cos	Shared Proteins	Shared Effects	Shared Drugs
Rank						
1	macrocytic anemia (disease) (disease_8672)	0.804716	0.952129	36	1	0
2	sideroblastic anemia (disease_15701)	0.694448	0.961711	1	2	0
3	pancytopenia (disease_3251)	0.660468	0.987846	3	0	0
4	beta-thalassemia and related diseases (disease_4814)	0.604658	0.759981	2	0	0
5	microcytic anemia (disease_1809)	0.593104	0.716523	0	1	0
6	hypochromic anemia (disease_2332)	0.591500	0.688182	0	1	0
7	undifferentiated carcinoma of the corpus uteri (disease_4026)	0.581832	0.950627	12	0	0
8	breast cancer (disease_14717)	0.572418	0.896834	12	0	0
9	aplastic anemia (disease_3742)	0.569642	0.648997	2	0	0
10	hepatocellular carcinoma (disease_14718)	0.569100	0.874473	14	0	0

OUTPUT 3 — EVIDENCE & SIMILARITY SUBGRAPH

Drug Repurposing Evidence Graph — anemia (disease) (model: HGT)
Drugs · Protein Evidence · Phenotypes · Similar Diseases · Safety



Subgraph saved → /mnt/data/demo_subgraph.png

OUTPUT 4 — RAG EXPLANATION & SUMMARY

Disease query : anemia (disease)
Model : HGT

SIMILAR DISEASES USED AS CONTEXT

Diseases most biologically similar to the query were identified using:
embedding cosine similarity · disease ontology edges · shared proteins
PPI-expanded protein bridges · phenotype overlap · shared therapeutics

- macrocytic anemia (disease) (disease_8672) (similarity=0.805, embedding=0.952, shared proteins=36, shared effects=1)
- sideroblastic anemia (disease_15701) (similarity=0.694, embedding=0.962, shared proteins=1, shared effects=2)
- pancytopenia (disease_3251) (similarity=0.660, embedding=0.988, shared proteins=3, shared effects=0)
- beta-thalassemia and related diseases (disease_4814) (similarity=0.605, embedding=0.760, shared proteins=2, shared effects=0)
- microcytic anemia (disease_1809) (similarity=0.593, embedding=0.717, shared proteins=0, shared effects=1)

TOP 5 CANDIDATE DRUGS

Rank 1 – Testosterone (drug_600)

Currently known to treat : premature menopause, combined pituitary hormone deficiencies, genetic form, pituitary deficiency (+20 more)
Raw model score : 0.566
Similar disease support : 0.000
Final rerank score : 0.512
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Protein interaction bridge: Testosterone (drug_600) --[drug_target_protein]--> ESR1 (protein_4237) --[protein_interacts_with_protein]--> GATA1 (protein_5939) <--[disease_associated_with_p
- Protein interaction bridge: Testosterone (drug_600) --[drug_target_protein]--> ESR1 (protein_4237) --[protein_interacts_with_protein]--> PRL (protein_12181) <--[disease_associated_with_pr
- Protein interaction bridge: Testosterone (drug_600) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> CDK6 (protein_535) <--[disease_associated_with_prote
- Protein interaction bridge: Testosterone (drug_600) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> CSF3 (protein_2809) <--[disease_associated_with_prot

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 5
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

No strong support identified from similar diseases.

Rank 2 – Testosterone enanthate (drug_6450)

Currently known to treat : premature menopause, trisomy 13, androgen insensitivity syndrome (+3 more)
Raw model score : 0.562
Similar disease support : 0.000
Final rerank score : 0.509
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Protein interaction bridge: Testosterone enanthate (drug_6450) --[drug_target_protein]--> ESR1 (protein_4237) --[protein_interacts_with_protein]--> GATA1 (protein_5939) <--[disease_associ
- Protein interaction bridge: Testosterone enanthate (drug_6450) --[drug_target_protein]--> ESR1 (protein_4237) --[protein_interacts_with_protein]--> PRL (protein_12181) <--[disease_associa
- Protein interaction bridge: Testosterone enanthate (drug_6450) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> CDK6 (protein_535) <--[disease_associated
- Protein interaction bridge: Testosterone enanthate (drug_6450) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> CSF3 (protein_2809) <--[disease_associate

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 5
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:
• No direct contraindication or adverse overlap signal found.

Support from similar diseases:
No strong support identified from similar diseases.

Rank 3 – Ketoconazole (drug_995)

Currently known to treat : chromomycosis, white piedra, Cushing syndrome (+11 more)
Raw model score : 0.561
Similar disease support : 0.000
Final rerank score : 0.509
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:
• Protein interaction bridge: Ketoconazole (drug_995) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> CDK6 (protein_535) <--[disease_associated_with_protein]--> HIV-1 infection (disease_8665)
• Protein interaction bridge: Ketoconazole (drug_995) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> CSF3 (protein_2809) <--[disease_associated_with_protein]--> HIV-1 infection (disease_8665)
• Protein interaction bridge: Ketoconazole (drug_995) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> THRA (protein_14783) <--[disease_associated_with_protein]--> HIV-1 infection (disease_8665)
• Protein interaction bridge: Ketoconazole (drug_995) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> TNFRSF10A (protein_17606) <--[disease_associated_with_protein]--> HIV-1 infection (disease_8665)

Evidence profile:
• Shared protein bridges : 0
• Protein interaction bridges : 4
• Disease-to-disease transfer : 0
• Phenotype / effect overlap : 0

Safety signals:
• No direct contraindication or adverse overlap signal found.

Support from similar diseases:
No strong support identified from similar diseases.

Rank 4 – Cisplatin (drug_494)

Currently known to treat : borderline epithelial tumor of ovary, malignant non-dysgerminomatous germ cell tumor of ovary, familial ovarian cancer (+12 more)
Raw model score : 0.561
Similar disease support : 0.000
Final rerank score : 0.509
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:
• Shared protein target: Cisplatin (drug_494) --[drug_transporter_protein]--> ATP7A (protein_10854) <--[disease_associated_with_protein]--> anemia (disease_8665)
• Protein interaction bridge: Cisplatin (drug_494) --[drug_enzyme_protein]--> SOD1 (protein_14364) --[protein_interacts_with_protein]--> SOD2 (protein_14367) <--[disease_associated_with_protein]--> HIV-1 infection (disease_8665)
• Protein interaction bridge: Cisplatin (drug_494) --[drug_target_protein]--> A2M (protein_4024) --[protein_interacts_with_protein]--> IL2 (protein_8037) <--[disease_associated_with_protein]--> HIV-1 infection (disease_8665)
• Protein interaction bridge: Cisplatin (drug_494) --[drug_target_protein]--> A2M (protein_4024) --[protein_interacts_with_protein]--> TNF (protein_14840) <--[disease_associated_with_protein]--> HIV-1 infection (disease_8665)

Evidence profile:
• Shared protein bridges : 1
• Protein interaction bridges : 5
• Disease-to-disease transfer : 0
• Phenotype / effect overlap : 0

Safety signals:
• No direct contraindication or adverse overlap signal found.

Support from similar diseases:
No strong support identified from similar diseases.

Rank 5 – Methotrexate (drug_540)

Currently known to treat : juvenile idiopathic arthritis, Langerhans cell histiocytosis, non-Hodgkin lymphoma (+15 more)
Raw model score : 0.562

Similar disease support : 0.000
Final rerank score : 0.434
Risk assessment : high observed graph-based safety concern

Mechanistic evidence:

- Shared protein target: Methotrexate (drug_540) --[drug_enzyme_protein]--> TYMS (protein_15065) <--[disease_associated_with_protein]-- anemia (disease) (disease_8665)
- Shared protein target: Methotrexate (drug_540) --[drug_transporter_protein]--> SLC46A1 (protein_1578) <--[disease_associated_with_protein]-- anemia (disease) (disease_8665)
- Shared protein target: Methotrexate (drug_540) --[drug_enzyme_protein]--> MTHFR (protein_9521) <--[disease_associated_with_protein]-- anemia (disease) (disease_8665)
- Protein interaction bridge: Methotrexate (drug_540) --[drug_enzyme_protein]--> ATIC (protein_9669) --[protein_interacts_with_protein]--> TYMS (protein_15065) <--[disease_associated_with_p

Evidence profile:

- Shared protein bridges : 3
- Protein interaction bridges : 2
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- Direct safety warning: Direct contraindication: Methotrexate (drug_540) --[contraindication]-- anemia (disease) (disease_8665)

Support from similar diseases:
No strong support identified from similar diseases.

INTERPRETATION NOTE

These are graph-based research hypotheses, not clinical prescriptions.
The strongest candidates combine: high model score + strong mechanistic evidence (especially shared protein bridges) + support from biologically similar diseases + no direct contraindication signals.

Always validate predictions against current clinical literature before drawing any experimental or clinical conclusions.

Structured RAG report table:

rank	drug	known_treatment_for	model_score	similar_disease_support	rerank_score	risk	evidence_summary	safety_summary
1	Testosterone (drug_600)	premature menopause, combined pituitary hormone deficiencies, genetic form, pituitary deficiency (+20 more)	0.566466	0.0	0.511556	low	Protein interaction bridge: Testosterone (drug_600) --[drug_target_protein]--> ESR1 (protein_4237) --[protein_intera...	No direct contraindication or adverse overlap signal found.
2	Testosterone enanthate (drug_6450)	premature menopause, trisomy 13, androgen insensitivity syndrome (+3 more)	0.562182	0.0	0.509200	low	Protein interaction bridge: Testosterone enanthate (drug_6450) --[drug_target_protein]--> ESR1 (protein_4237) --[pro...	No direct contraindication or adverse overlap signal found.
3	Ketoconazole (drug_995)	chromomycosis, white piedra, Cushing syndrome (+11 more)	0.561272	0.0	0.508700	low	Protein interaction bridge: Ketoconazole (drug_995) --[drug_target_protein]--> AR (protein_8143) --[protein_interact...	No direct contraindication or adverse overlap signal found.
4	Cisplatin (drug_494)	borderline epithelial tumor of ovary, malignant non-dysgerminomatous germ cell tumor of ovary, familial ovarian canc...	0.561239	0.0	0.508682	low	Shared protein target: Cisplatin (drug_494) --[drug_transporter_protein]--> ATP7A (protein_10854) <--[disease_associ...	No direct contraindication or adverse overlap signal found.
5	Methotrexate (drug_540)	juvenile idiopathic arthritis, Langerhans cell histiocytosis, non-Hodgkin lymphoma (+15 more)	0.562198	0.0	0.434209	high	Shared protein target: Methotrexate (drug_540) --[drug_enzyme_protein]--> TYMS (protein_15065) <--[disease_associate...	Direct safety warning: Direct contraindication: Methotrexate (drug_540) --[contraindication]-- anemia (disease) (dis...